Vendor Sales Performance Analysis

1. Project Overview

This project analyses the purchasing, sales and inventory data of a multi-store liquor and wine retail business to understand which vendors and products drive revenue and profit, which ones tie up capital and where costs can be reduced.

I built the full pipeline myself, raw csv files were ingested into a SQL database, combined into a single vendor-level summary table with SQL, cleaned, enriched and analyzed research questions with Python (pandas) and finally presented in an interactive Power BI dashboard.

Tools used: Python (pandas, SQLAlchemy, sqlite3, logging), SQL (SQLite, CTEs, joins, aggregations), Power BI, Jupyter Notebook

2. Business Problem

A retailer sells goods to many vendors for dozens of stores. The data is distributed into multiple tables such as purchase price, selling price, invoice and freight data sit in separate tables, so it is hard to answer basic questions such as:

- Which vendors generate the most sales and how dependent is the business on a few of them?
- Which brands earn high margins but sell too little?
- Which vendors have slow-moving stock, so money is stuck in unsold inventory?
- How do freight and excise tax affect profitability?

3. Dataset

Data was distributed in 6 separate csv files.

a. **Purchases Table-** contains information about purchases done by every vendor for their stores. It contains data like purchase quantity, purchase price, payment in dollars, name of product.

b. **Purchase Price-** contains information about the selling price and size of each product.
c. **Sales -** contains data of sales done, quantity sold and taxes paid by vendor for each order.
d. **Vendor Invoice -** Information about purchases of vendors, it has quantity bought by vendor, total payment in dollars by vendors, Fright cost etc. Fright is extra charges such as taxes, exporting charges paid by the company.
e. **Begin Inventory and End inventory tables** - these tables do not serve our purpose.

For analysis, I created SQLite database "inventory.db" using sqlalchemy library. I wrote a python script to store all csv files separately in the inventory database.

Extracted all useful information from all these 6 files and created a summary table.

4. Data Preprocessing

- I wrote a Python script that uses the SQLAlchemy library to load the raw CSV files into a SQLite database. I then connected to the database and read the tables into pandas for exploration.
- Using SQL, I extracted the relevant fields from each table, joined them and built a consolidated table named `vendor_sales_summary`.
- Finally, I cleaned the data in Python by converting data types, filling missing values and removing inconsistencies such as extra spaces in vendor names.

5. EDA Insights

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df.describe().T
```

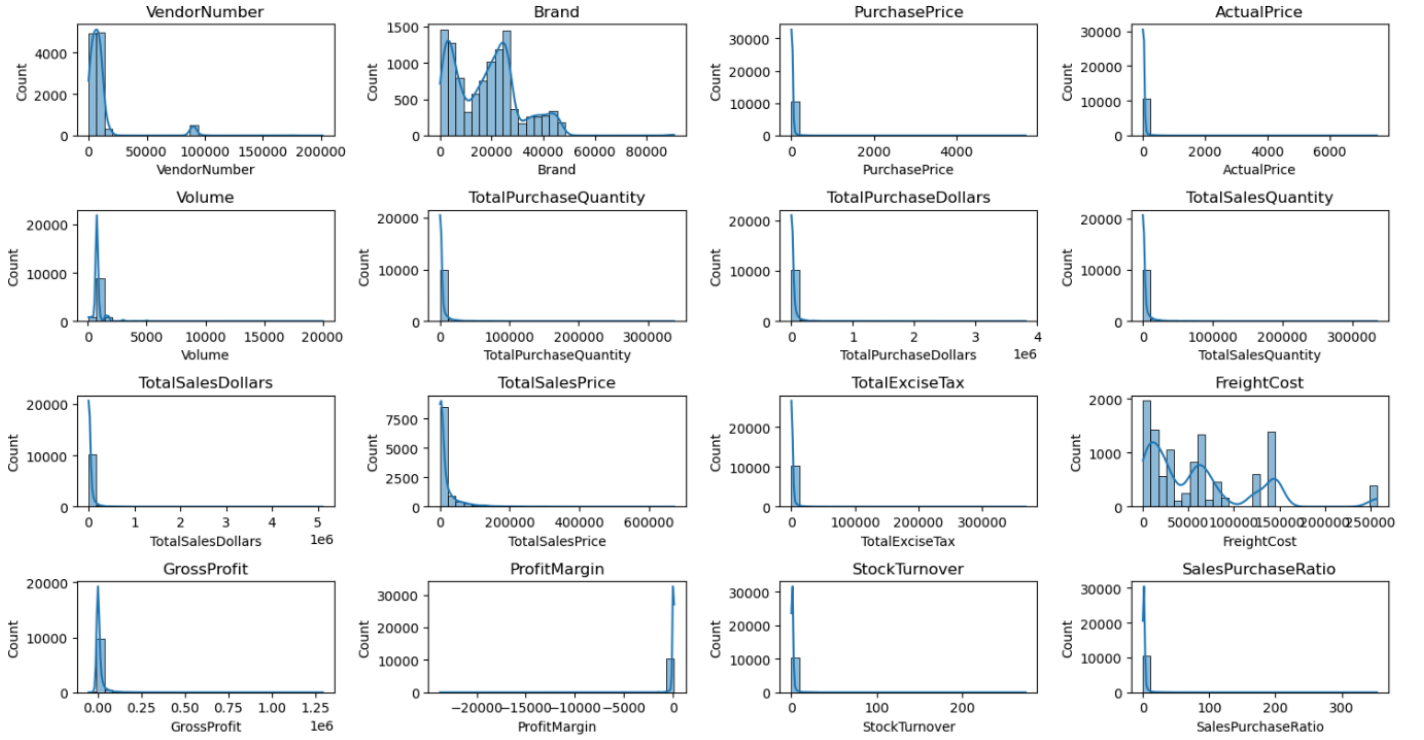
	count	mean	std	min	25%	50%	75%	max
VendorNumber	10692.0	1.065065e+04	18753.519148	2.00	3951.000000	7153.000000	9552.000000	2.013590e+05
Brand	10692.0	1.803923e+04	12662.187074	58.00	5793.500000	18761.500000	25514.250000	9.063100e+04
PurchasePrice	10692.0	2.438530e+01	109.269375	0.36	6.840000	10.455000	19.482500	5.681810e+03
ActualPrice	10692.0	3.564367e+01	148.246016	0.49	10.990000	15.990000	28.990000	7.499990e+03
Volume	10692.0	8.473605e+02	664.309212	50.00	750.000000	750.000000	750.000000	2.000000e+04
TotalPurchaseQuantity	10692.0	3.140887e+03	11095.086769	1.00	36.000000	262.000000	1975.750000	3.376600e+05
TotalPurchaseDollars	10692.0	3.010669e+04	123067.799627	0.71	453.457500	3655.465000	20738.245000	3.811252e+06
TotalSalesQuantity	10692.0	3.077482e+03	10952.851391	0.00	33.000000	261.000000	1929.250000	3.349390e+05
TotalSalesDollars	10692.0	4.223907e+04	167655.265984	0.00	729.220000	5298.045000	28396.915000	5.101920e+06
TotalSalesPrice	10692.0	1.879378e+04	44952.773386	0.00	289.710000	2857.800000	16059.562500	6.728193e+05
TotalExciseTax	10692.0	1.774226e+03	10975.582240	0.00	4.800000	46.570000	418.650000	3.682428e+05
FreightCost	10692.0	6.143376e+04	60938.458032	0.09	14069.870000	50293.620000	79528.990000	2.570321e+05
GrossProfit	10692.0	1.213238e+04	46224.337964	-52002.78	52.920000	1399.640000	8660.200000	1.290668e+06
ProfitMargin	10692.0	-inf	NaN	-inf	13.324515	30.405457	39.956135	9.971666e+01
StockTurnover	10692.0	1.706793e+00	6.020460	0.00	0.807229	0.981529	1.039342	2.745000e+02
SalesPurchaseRatio	10692.0	2.504390e+00	8.459067	0.00	1.153729	1.436894	1.665449	3.529286e+02

Negative & Zero Values:

Gross Profit: Minimum value is -52,002.78, indicating losses.

Profit Margin: Has a minimum value of negative infinity, which is due to in some places gross profit is negative means loss and at those places TotalSalesDollars is 0.

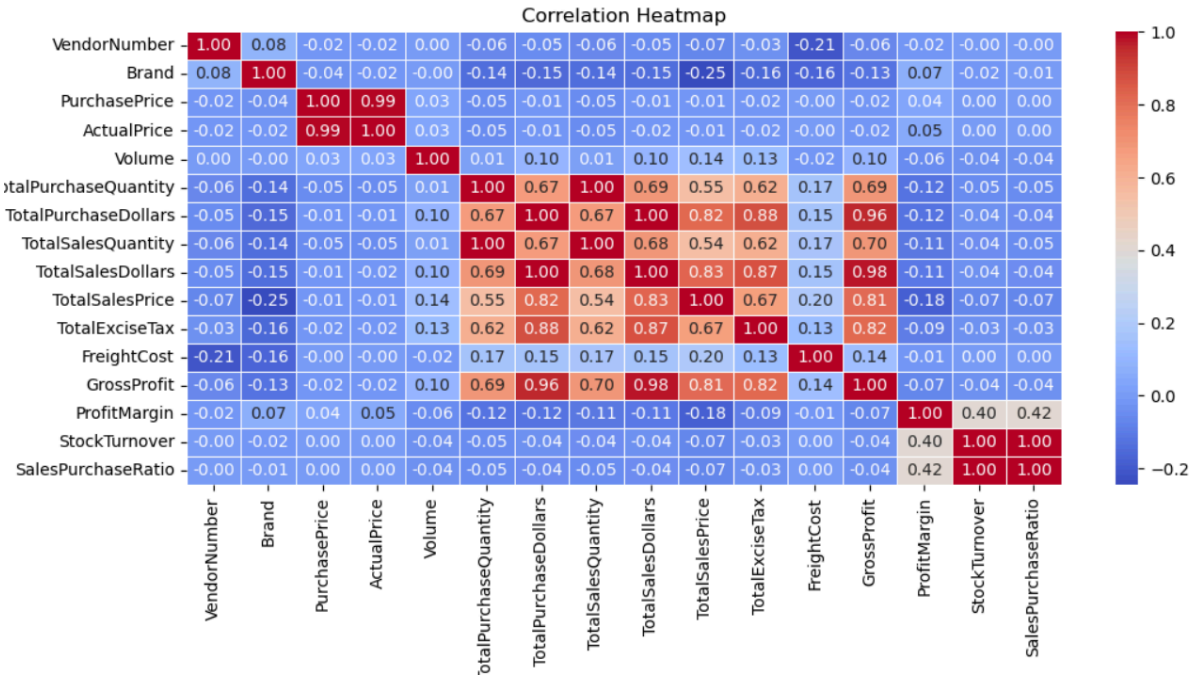
Total Sales Quantity & Sales Dollars: Minimum values are 0, meaning some products were purchased but never sold. These could be slow-moving or obsolete stock.



Outliers Indicated by High Standard Deviations:

- **Purchase & Actual Prices:** The max values (5,681.81 & 7,499.99) are significantly higher than the mean (24.39 & 35.64) respectively, that could be due to potential premium products.
- **Freight Cost:** Huge variation, from 0.09 to 257,032.07, suggests logistics inefficiencies or bulk shipments.
- **Stock Turnover:** Ranges from 0 to 274.5, implying some products sell extremely fast while others remain in stock indefinitely. Value more than 1 indicates that Sold quantity for that product is higher than purchased quantity due to either sales being fulfilled from older stock.

Correlation Insights



- PurchasePrice has weak correlations with TotalSalesDollars (-0.012) and GrossProfit (-0.016), suggesting that price variations do not significantly impact sales revenue or profit.
- Strong correlation between total purchase quantity and total sales quantity (0.999), confirming efficient inventory turnover.
- StockTurnover has weak negative correlations with both GrossProfit (-0.038) and ProfitMargin (-0.055), indicating that faster turnover does not necessarily result in higher profitability.

6. Data Filtering

To enhance the reliability of the insights, we removed inconsistent data points where:

- Gross Profit ≤ 0 (to exclude transactions leading to losses)
- Profit Margin ≤ 0 (to ensure analysis focuses on profitable transactions)
- Total Sales Quantity = 0 (to eliminate inventory that was never sold)

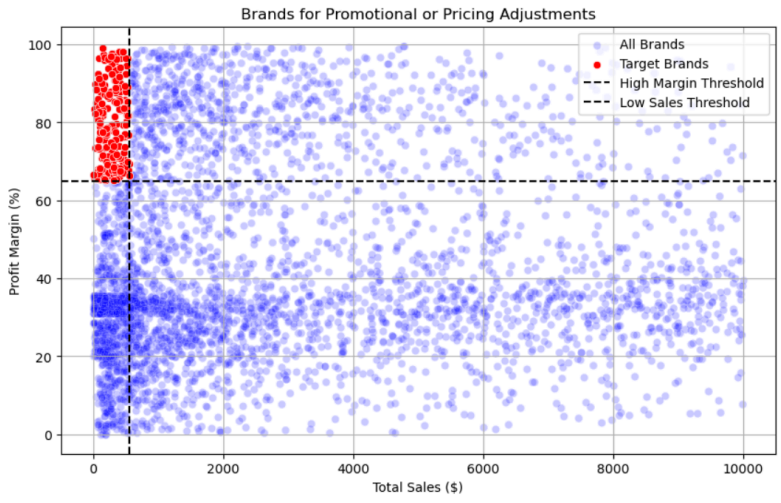
6. Research question and Key findings

a. Brand for Promotional and Pricing Adjustments which exhibit lower sales performance but higher profit margins

198 brands exhibit lower sales but higher profit margins, which could benefit from targeted marketing, promotions, or price optimizations to increase volume without compromising profitability.

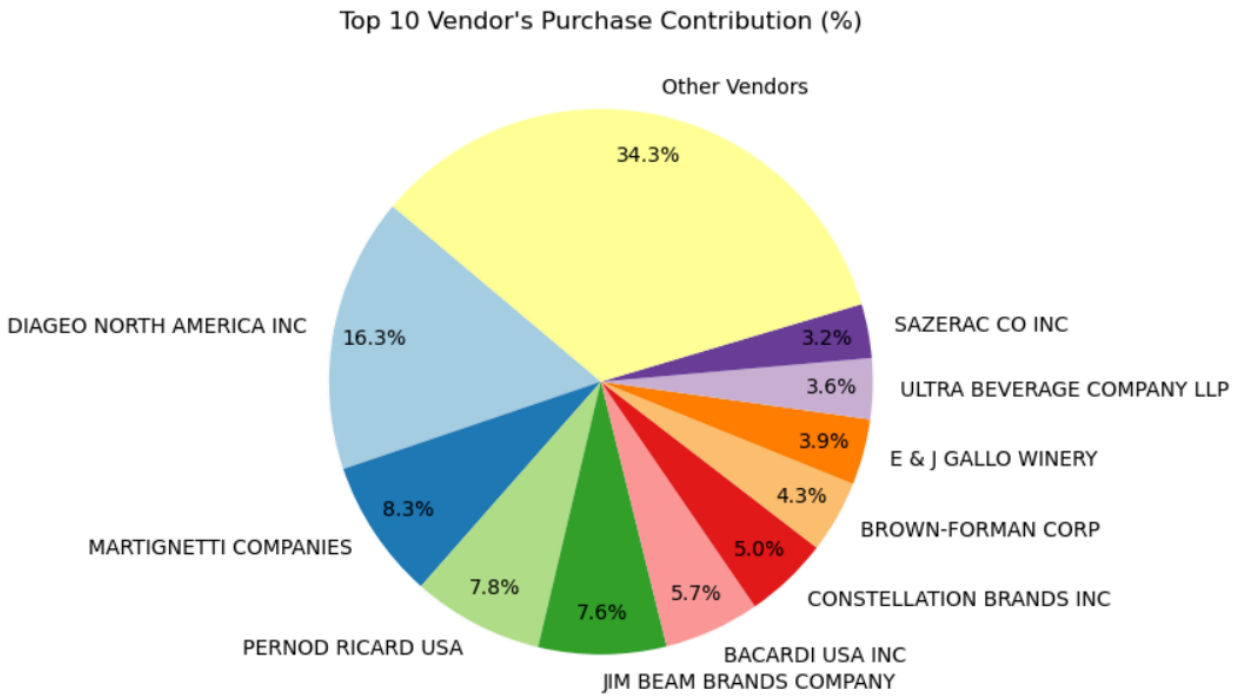
	Description	TotalSalesDollars	ProfitMargin
6199	Santa Rita Organic Svgn Bl	9.99	66.466466
2369	Debauchery Pnt Nr	11.58	65.975820
2070	Concannon Glen Ellen Wh Zin	15.95	83.448276
2188	Crown Royal Apple	27.86	89.806174
6237	Sauza Sprklg Wild Berry Marg	27.96	82.153076
...	...	...	...
5074	Nanbu Bijin Southern Beauty	535.68	76.747312
2271	Dad's Hat Rye Whiskey	538.89	81.851584
57	A Bichot Clos Marechaudes	539.94	67.740860
6245	Sbragia Home Ranch Merlot	549.75	66.444748
3326	Goulee Cos d'Estournel 10	558.87	69.434752

198 rows × 3 columns



b. Top Vendors by Sales & Purchase Contribution

The top 10 vendors contribute 65.69% of total purchases, while the remaining vendors contribute only 34.31%. This over-reliance on a few vendors may introduce risks such as supply chain disruptions, indicating a need for diversification.



c. Impact of Bulk Purchasing on Cost Savings

- Vendors buying in large quantities receive a 72% lower unit cost ($10.78 per unit vs. higher unit costs in smaller orders).
- Bulk pricing strategies encourage larger orders, increasing total sales while maintaining profitability.
- If a vendor buys 85 to 1500 units in bulk which is the optimal purchase of volume for cost saving, as he can get product at avg price of 15.5$.

OrderSize	UnitPurchasePrice
Small	39.068186
Medium	15.486414
Large	10.777625

d. Identifying Vendors with Low Inventory Turnover

- Total Unsold Inventory Capital: $2.71M
- Slow-moving inventory increases storage costs, reduces cash flow efficiency and affects overall profitability.
- Identifying vendors with low inventory turnover enables better stock management, minimizing financial strain.

	VendorName	UnsoldInventoryValue
25	DIAGEO NORTH AMERICA INC	722.21K
46	JIM BEAM BRANDS COMPANY	554.67K
68	PERNOD RICARD USA	470.63K
116	WILLIAM GRANT & SONS INC	401.96K
30	E & J GALLO WINERY	228.28K
79	SAZERAC CO INC	198.44K
11	BROWN-FORMAN CORP	177.73K
20	CONSTELLATION BRANDS INC	133.62K
61	MOET HENNESSY USA INC	126.48K
77	REMY COINTREAU USA INC	118.60K

VendorName	StockTurnover
ALISA CARR BEVERAGES	0.615385
HIGHLAND WINE MERCHANTS LLC	0.708333
PARK STREET IMPORTS LLC	0.751306
Circa Wines	0.755676
Dunn Wine Brokers	0.766022
CENTEUR IMPORTS LLC	0.773953
SMOKY QUARTZ DISTILLERY LLC	0.783835
TAMWORTH DISTILLING	0.797078
THE IMPORTED GRAPE LLC	0.807569
WALPOLE MTN VIEW WINERY	0.820548

e. Profit Margin Comparison: High vs. Low-Performing Vendors

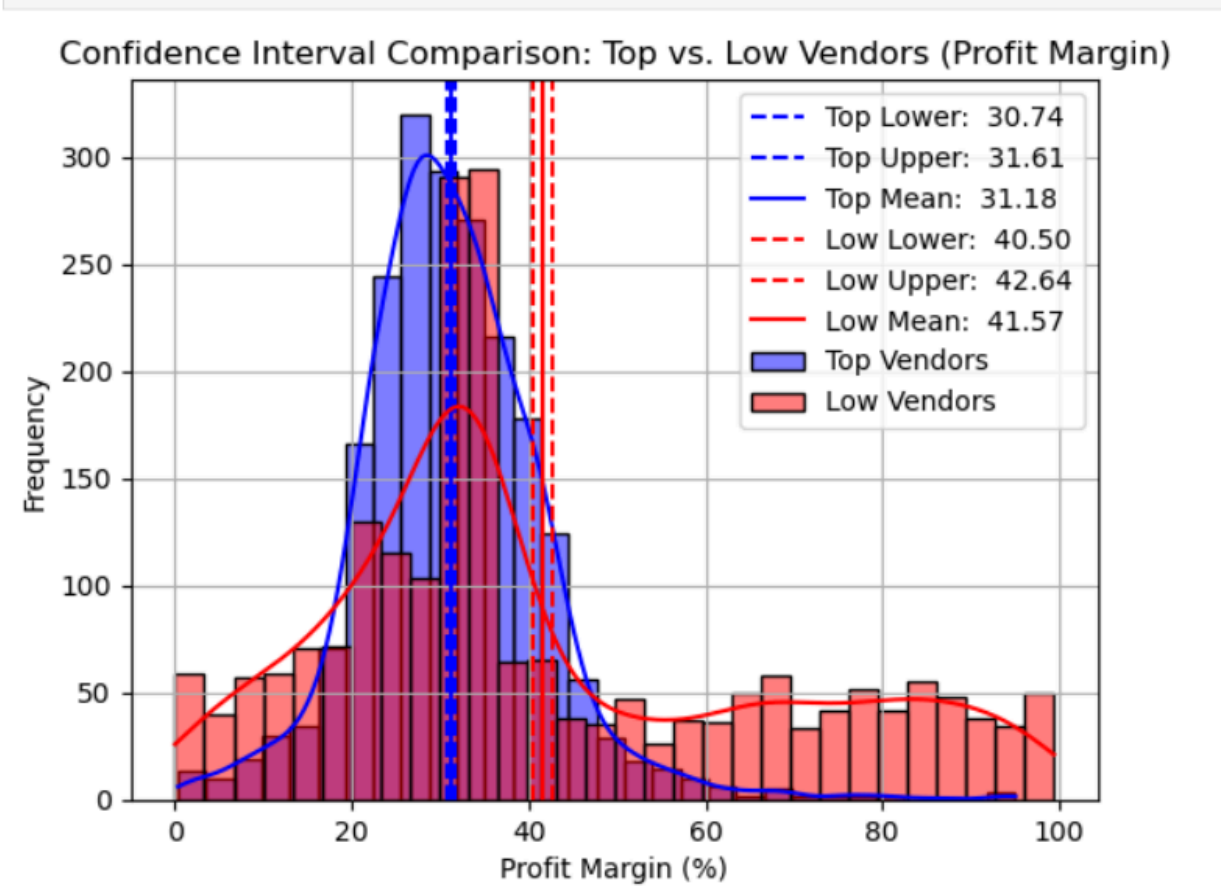
Top Vendors' Profit Margin (95% CI): (30.74%, 31.61%), Mean: 31.17%

Low Vendors' Profit Margin (95% CI): (40.48%, 42.62%), Mean: 41.55%

Low-performing vendors maintain higher margins but struggle with sales volumes, indicating potential pricing inefficiencies or market reach issues.

Actionable insights

- Top-performing vendors: Optimize profitability by adjusting pricing, reducing operational costs or offering bundled promotions.
- Low-performing vendors: Improve marketing efforts, optimize pricing strategies and enhance distribution networks.



f. Statistical Validation of Profit Margin Differences

Hypothesis Testing:

Ho (Null Hypothesis): No significant difference in profit margins between top and low-performing vendors.

H1 (Alternative Hypothesis): A significant difference exists in profit margins between the two vendor groups.

Result: The null hypothesis is rejected, confirming that the two groups operate under distinctly different profitability models.

Implication: High-margin vendors may benefit from better pricing strategies, while top-selling vendors could focus on cost efficiency.

7. Final Recommendations

- Re-evaluate pricing for low-sales, high-margin brands to boost sales volume without sacrificing profitability.
- Diversify vendor partnerships to reduce dependency on a few suppliers and mitigate supply chain risks.
- Leverage bulk purchasing advantages to maintain competitive pricing while optimizing inventory management.
- Optimize slow-moving inventory by adjusting purchase quantities, launching clearance sales or revising storage strategies.
- Enhance marketing and distribution strategies for low-performing vendors to drive higher sales volumes without compromising profit margins.
- By implementing these recommendations, the company can achieve
- sustainable profitability, mitigate risks and enhance overall Operational efficiency.